Assessment Growth and Section Performance Analytics in R

Public-Safe BOY/EOY Improvement Study

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Recommendation

This study evaluates **beginning-of-year to end-of-year improvement** in a public-safe education assessment extract. The business question is: which course sections improved more or less than expected after accounting for starting performance, readiness, attendance, course track, grade level, and school-year context?

Use BOY/EOY gain as the headline metric, but use **adjusted growth signals** for section review. Raw gains are easy to understand, but they can reward sections that started low and penalize sections that started near a ceiling. The adjusted signal compares observed gain with expected gain for a similar starting profile.

The analysis includes 1,737 paired BOY/EOY records across 174 section-year groups and 5 simulated teachers. The average BOY score is 48.5, the average EOY score is 54.2, and the mean raw gain is 5.72 points.

The selected expected-growth model is **Readiness-augmented**. It achieved holdout RMSE **4.399** and holdout R-squared **0.181**. Section signals should be used for instructional review, curriculum support, and follow-up analysis; they should not be used as automatic teacher evaluation or personnel decisions.

Direct Answers

1. The main metric is BOY/EOY score improvement: end-of-year score minus beginning-of-year score for the same simulated student in the same section and teacher context.
2. The average raw gain is 5.72 points across 1,737 paired records.
3. Raw section gains are reported, but the primary comparison is adjusted growth: observed section gain minus expected gain from the selected model.
4. The section review layer flags 8 section-year groups above expected growth and 5 below expected growth, with 156 within expected range.
5. The teacher and course summaries are aggregation views for leadership conversations. They are not personnel ratings because the data is public-safe, simulated, and section composition still matters.

Data Audit

The analysis starts from a public-safe assessment extract. A record enters the growth model only when the same simulated student has valid BOY and EOY scores in the same section and with the same simulated teacher. This keeps the improvement metric tied to a section experience instead of mixing students across sections.

Measure	Value
Raw assessment rows	4,018
BOY/EOY candidate pairs	2,009
Included paired records	1,737
Unique public-safe student IDs	671
Unique section-year groups	174
Unique simulated teachers	5
Mean BOY score	48.5
Mean EOY score	54.2
Mean BOY/EOY gain	5.7
Median section paired records	10

The extract uses simulated identifiers and generalized score/readiness behavior from a bootstrapped assessment workflow. It is not a release of real student records or real personnel data.

Raw Section Improvement

The first layer is descriptive: calculate the BOY/EOY score gain inside each section-year group and run a paired-improvement t-test against zero. This answers whether a section improved, but it does not by itself prove that the section improved more than expected given its starting point.

The table below shows high-signal section-year groups from the review layer, with their raw BOY/EOY t-test results included for context. The full section t-test table is generated as `reports/section_ttests.csv`.

Section	N	BOY	EOY	Gain	95% CI	p-value
30-31 SEC-006	9	42.4	52.8	10.39	7.18 to 13.60	<0.001
30-31 SEC-019	12	37.6	46.7	9.09	6.66 to 11.52	<0.001
27-28 SEC-017	11	43.9	52.6	8.79	5.37 to 12.22	<0.001
29-30 SEC-017	11	37.1	45.9	8.77	5.77 to 11.78	<0.001
30-31 SEC-008	11	35.0	44.0	9.07	4.80 to 13.34	<0.001
31-32 SEC-002	10	40.1	49.1	9.02	6.32 to 11.73	<0.001
29-30 SEC-003	12	41.0	49.7	8.73	6.24 to 11.22	<0.001
26-27 SEC-006	9	30.0	39.9	9.85	6.41 to 13.28	<0.001

Adjusted Growth Model

The adjusted model estimates expected BOY/EOY gain from starting score/readiness and context. This is the key step that makes the analysis more useful than a raw gain ranking: it accounts for floor effects, ceiling effects, attendance context, course track, grade level, and school-year timing.

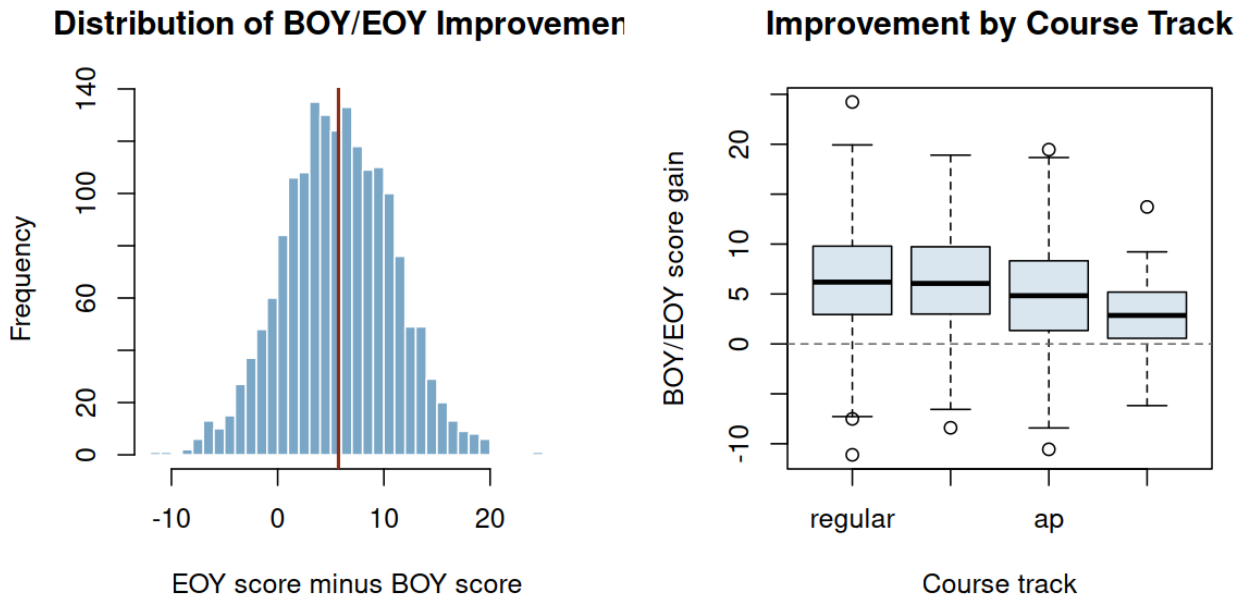

Figure 1: Distribution of BOY/EOY improvement

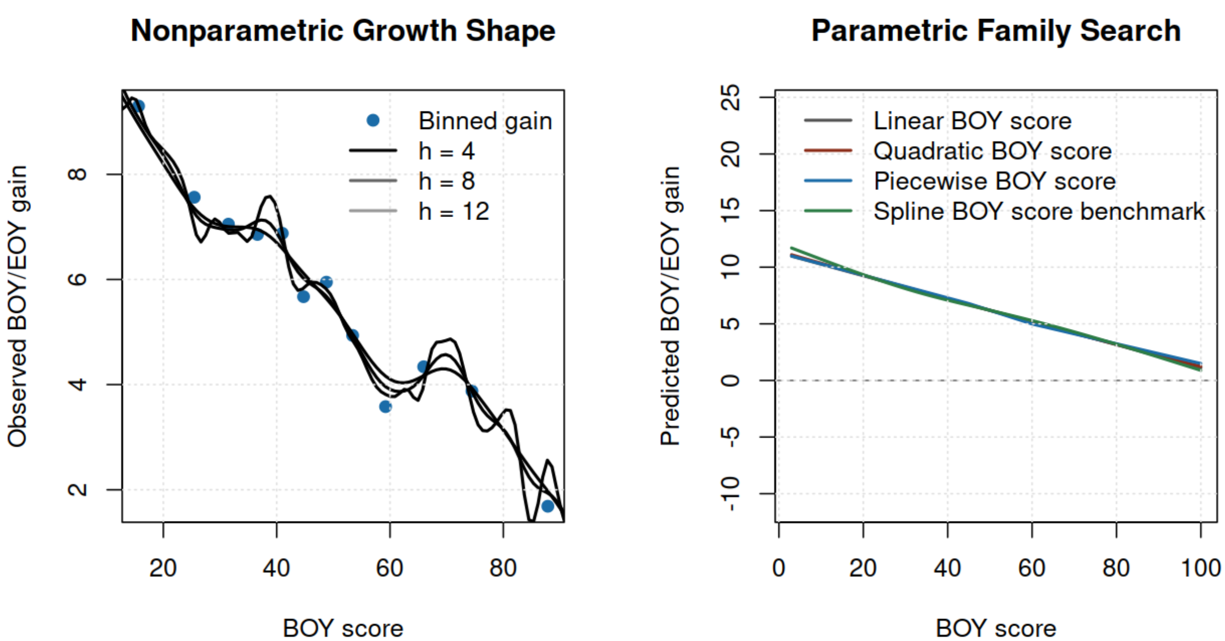

Figure 2: Nonparametric and parametric BOY score shape

Family	Why tested	Decision	CV RMSE	Holdout RMSE
Context baseline	Tests whether grade, course, attendance, and year context are enough.	Baseline comparator.	5.078	4.802
Linear BOY score	Adds the main baseline achievement signal.	Useful simple challenger.	4.734	4.428
Quadratic BOY score	Tests whether gain changes nonlinearly for very low or high BOY scores.	Compared for nonlinear gain shape.	4.737	4.427
Piecewise BOY score	Uses interpretable score regions to handle floor and ceiling effects.	Interpretable nonlinear challenger.	4.741	4.432
Readiness-augmented	Checks whether readiness adds signal beyond the observed BOY score.	Selected operating model.	4.700	4.399
Spline BOY score benchmark	Flexible benchmark for the baseline-score curve.	Benchmark only; not selected unless it materially improves validation.	4.745	4.422

Model	Selected	Role	Params	CV RMSE	CV SD	CV MAE	CV R2	Holdout RMSE	Holdout R2	Delta
Readiness-augmented	Yes	Selection candidate	13	4.700	0.010	3.773	0.159	4.399	0.181	0.000
Linear BOY score		Selection candidate	12	4.734	0.009	3.782	0.147	4.428	0.170	0.034
Quadratic BOY score		Selection candidate	13	4.737	0.010	3.786	0.146	4.427	0.171	0.037
Piecewise BOY score		Selection candidate	14	4.741	0.010	3.789	0.145	4.432	0.169	0.041

Model	Selected	Role	Params	CV RMSE	CV SD	CV MAE	CV R2	Holdout RMSE	Holdout R2	Delta
Spline BOY score benchmark		Benchmark	15	4.745	0.010	3.792	0.143	4.422	0.172	0.045
Context baseline		Selection candidate	10	5.078	0.007	4.076	0.019	4.802	0.024	0.378

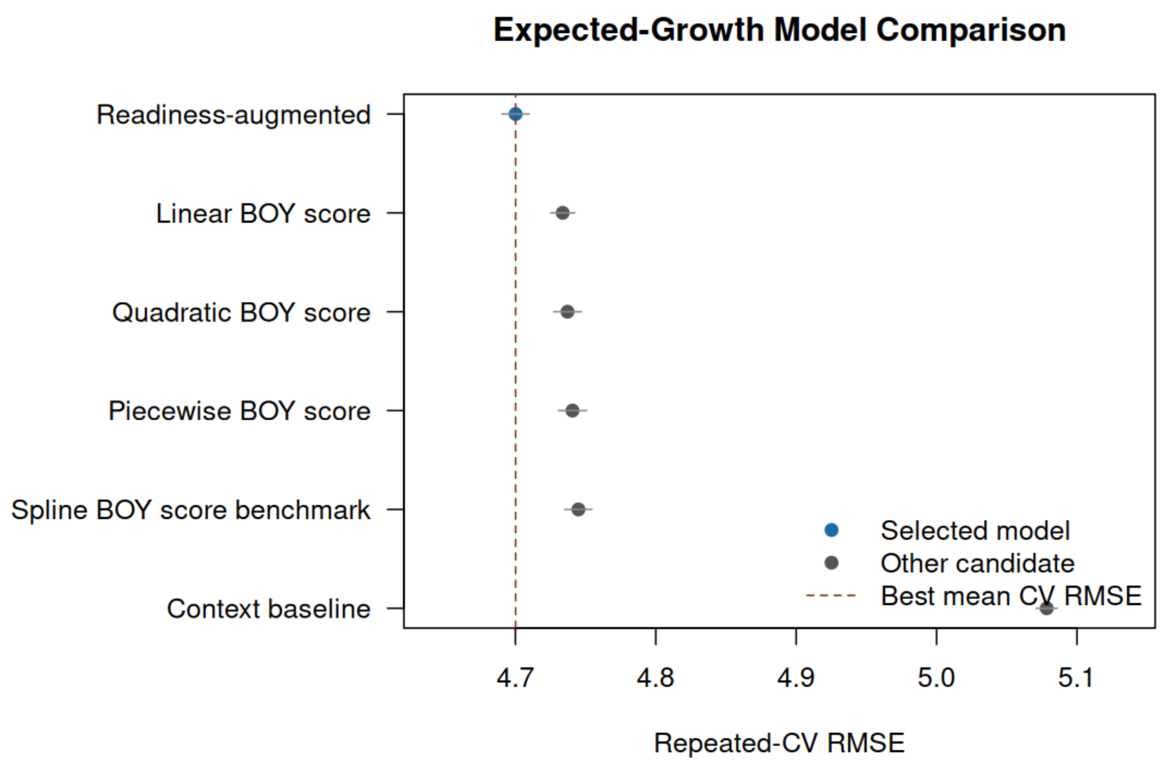

Figure 3: Expected-growth model comparison

Section Performance Signals

For each section-year group, the adjusted signal is the reliability-weighted average residual: observed gain minus expected gain, weighted toward zero for smaller groups. Positive values mean the section improved more than expected for its starting mix; negative values mean it improved less than expected.

Section	Teacher	N	Raw gain	Expected gain	Adjusted signal	Category
30-31 SEC-006	TCH-002	9	10.39	6.39	1.89	Above expected

Section	Teacher	N	Raw gain	Expected gain	Adjusted signal	Category
30-31 SEC-019	TCH-005	12	9.09	6.13	1.61	Above expected
27-28 SEC-017	TCH-005	11	8.79	5.77	1.59	Within expected range
29-30 SEC-017	TCH-003	11	8.77	5.92	1.50	Above expected
30-31 SEC-008	TCH-002	11	9.07	6.42	1.39	Within expected range
31-32 SEC-002	TCH-002	10	9.02	6.33	1.35	Within expected range
29-30 SEC-003	TCH-002	12	8.73	6.38	1.28	Above expected
26-27 SEC-006	TCH-002	9	9.85	7.18	1.26	Within expected range
30-31 SEC-009	TCH-002	9	2.33	6.65	-2.05	Below expected
30-31 SEC-002	TCH-002	12	2.22	5.48	-1.78	Within expected range
26-27 SEC-003	TCH-001	9	3.27	6.45	-1.50	Below expected
27-28 SEC-008	TCH-002	13	4.04	6.69	-1.50	Below expected
26-27 SEC-013	TCH-004	10	2.27	5.25	-1.49	Below expected
25-26 SEC-013	TCH-003	11	3.99	6.64	-1.39	Within expected range
31-32 SEC-019	TCH-005	12	2.41	4.90	-1.36	Within expected range
30-31 SEC-001	TCH-001	10	3.28	5.96	-1.34	Below expected

The full section signal table is generated as `reports/section_adjusted_signals.csv` so reviewers can inspect all section-year groups, not only the highlights shown in the report.

Instructor and Course Summary

Teacher and course summaries aggregate the section-level evidence. They are useful for spotting patterns that deserve follow-up, such as a course sequence that may need curriculum review or a teacher group whose sections repeatedly exceed expected growth. They should not be treated as standalone teacher

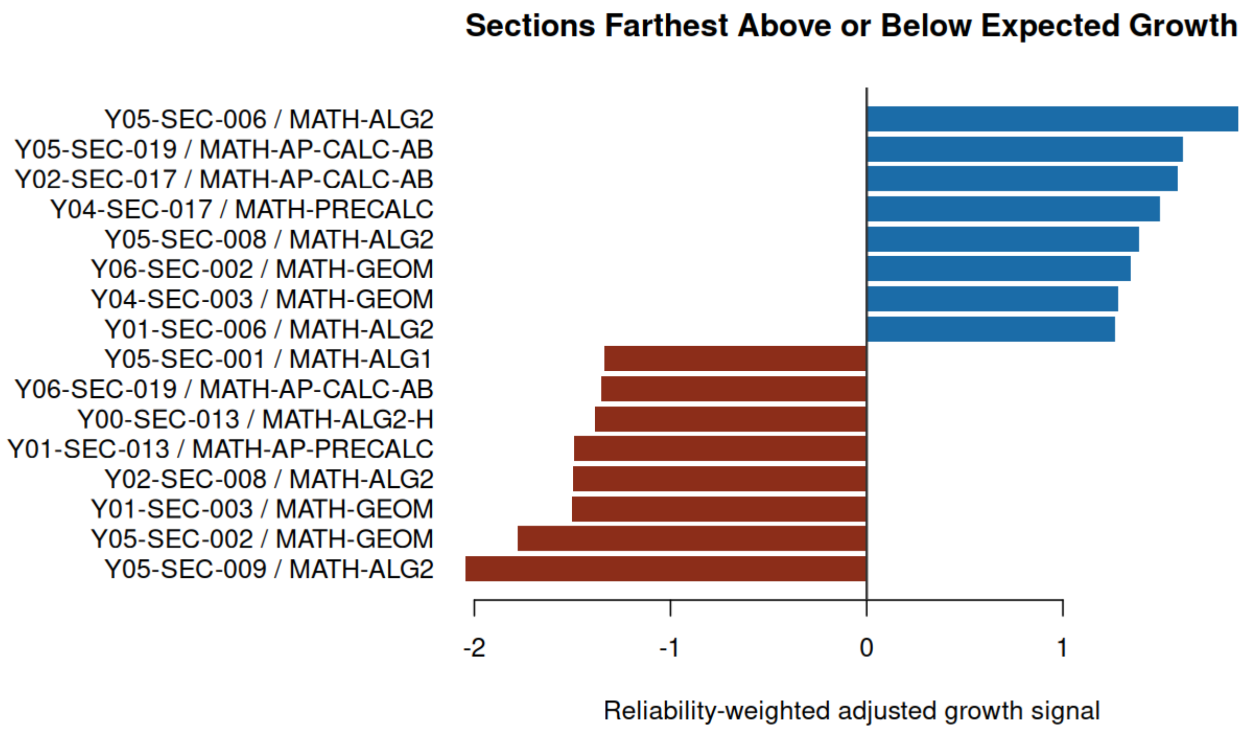

Figure 4: Sections above or below expected growth

quality scores.

Teacher	Sections	Records	BOY	EOY	Raw gain	Expected gain	Adjusted signal
TCH-005	41	420	54.6	59.4	4.81	4.77	0.04
TCH-002	40	424	39.6	46.1	6.48	6.45	0.03
TCH-004	36	313	55.8	60.9	5.11	5.10	0.01
TCH-003	34	351	50.6	56.5	5.95	5.94	0.01
TCH-001	23	229	40.7	47.2	6.49	6.66	-0.17

Course	Track	Sections	Records	BOY	EOY	Raw gain	Expected gain	Adjusted signal
MATH-PRECALC	regular	21	216	41.5	47.2	5.73	5.63	0.10
MATH-AP-CALC-AB	ap	27	294	50.5	55.7	5.21	5.14	0.07
MATH-ALG2	regular	23	236	37.2	43.8	6.57	6.52	0.05

Course	Track	Sections	Records	BOY	EOY	Raw gain	Expected gain	Adjusted signal
MATH-BEYOND-CORE	beyond_core	6	21	67.7	70.8	3.08	3.08	0.00
MATH-ALG2-H	honors	20	198	57.6	63.8	6.12	6.12	0.00
MATH-GEOM	regular	31	334	42.5	48.9	6.39	6.40	-0.00
MATH-AP-PRECALC	ap	22	214	59.2	64.3	5.10	5.14	-0.04
MATH-AP-CALC-BC	ap	14	131	63.7	67.5	3.86	3.97	-0.11
MATH-ALG1	regular	10	93	38.7	45.3	6.60	6.94	-0.34

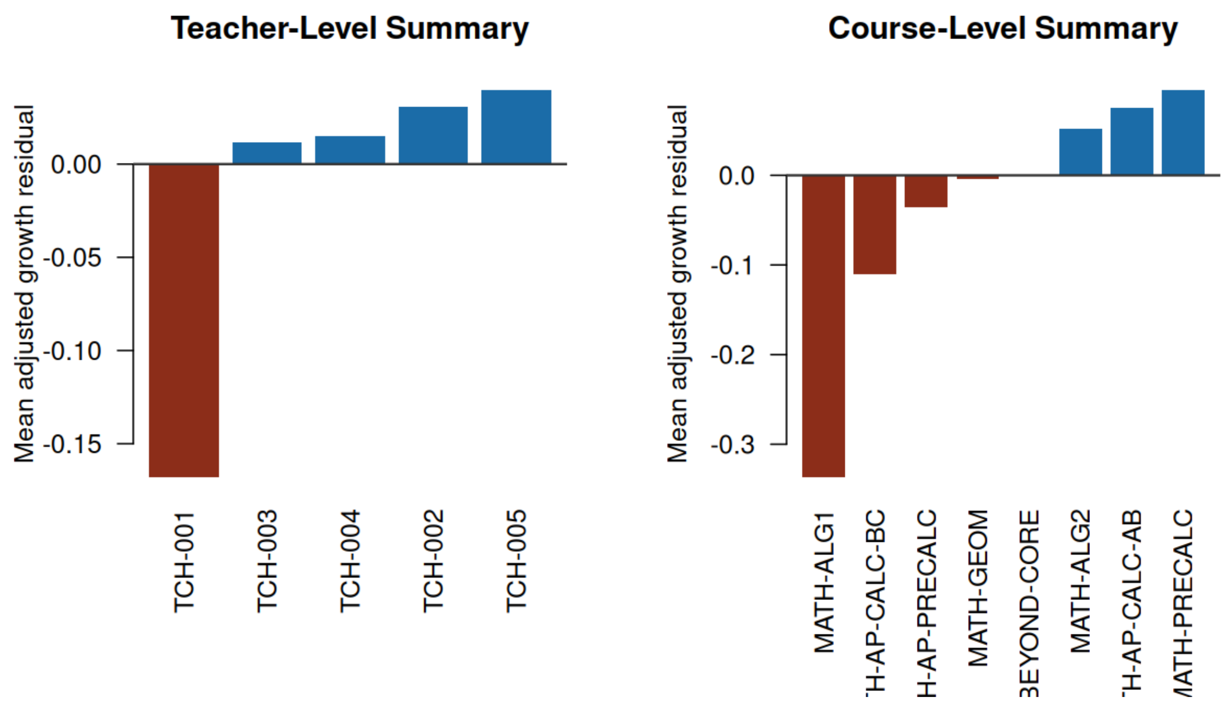

Figure 5: Teacher and course growth summaries

Diagnostics and Sensitivity

The diagnostics below check whether the expected-growth model is centered and whether the raw section rankings materially differ from adjusted rankings.

Diagnostic	Estimate	Interpretation
Holdout RMSE	4.399	Typical holdout prediction error on BOY/EOY gain
Holdout MAE	3.512	Average absolute holdout prediction error
Holdout R-squared	0.181	Share of holdout gain variation explained by the model
Residual mean, all pairs	0.000	Near 0 means expected gain is centered overall
Residual SD, all pairs	4.599	Residual spread used to judge section signal uncertainty

Measure	Value
Included paired records	1,737
Section-year groups	174
Groups with at least 5 paired records	169
Groups with at least 8 paired records	159
Mean raw BOY/EOY gain	5.72
Raw-vs-adjusted rank correlation	0.797
Top-10 overlap, raw vs adjusted ranking	70.0%

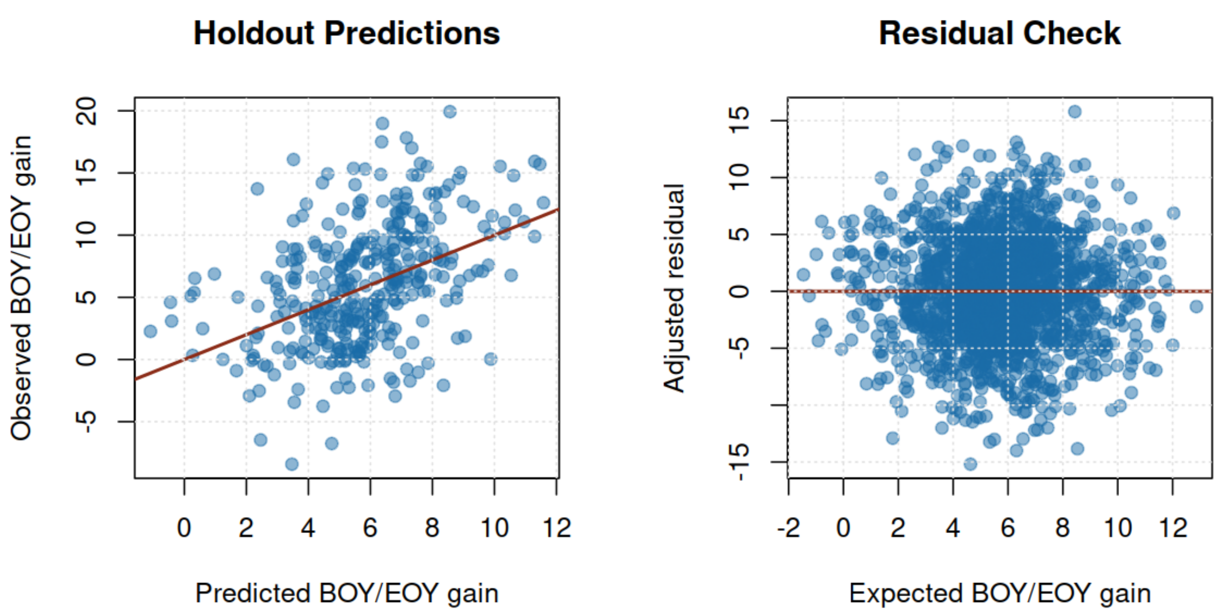

Figure 6: Growth model diagnostics

Bottom Line

- BOY/EOY improvement is the right stakeholder metric because it is easy to explain and aligned with instructional growth.

- Raw improvement should be shown, but adjusted growth should drive section review because starting level, attendance, course track, and ceiling effects matter.
- The strongest use case is instructional review: identify sections that exceed expected growth, sections that lag expected growth, and course patterns that deserve support.
- Avoid framing results as teacher quality rankings. The outputs are public-safe analytical signals, not personnel decisions.

Reproducibility

Rebuild the full evidence packet with `make all`. The core pipeline uses base R, the included public-safe extract, and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, real student-identifiable records, real personnel records, credentials, and copyrighted source documents.